



LECTURE NOTE- AGT 225: BEEF AND DAIRY



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1.0 SELECTION AND BREEDING

Selection and breeding form the genetic foundation of every successful beef and dairy enterprise. While good feeding, housing, and health management determine how well an animal expresses its genetic potential, selection and breeding determine what that potential actually is. A herd that is well-fed but poorly selected will never perform as well as a well-selected herd under the same management, which is why this topic opens the course.

1.1 Importance of Selection

Selection is the deliberate process of choosing which animals are allowed to breed and pass on their genes to the next generation. Farmers select for two broad categories of traits.

- Economic traits are those that directly affect the profitability of the enterprise. In dairy cattle, milk yield is the single most important economic trait, since it determines the farmer's primary revenue stream; in beef cattle, growth rate (weight gained per day) determines how quickly an animal reaches market weight. Fertility is economically important in both systems because a cow that fails to conceive or calve regularly produces no calf and, in dairy, no new lactation.
- Non-economic traits do not directly generate income but may still influence management decisions. Temperament affects how easily an animal can be handled, milked, or moved, with calmer animals generally being safer and less stressful to manage. Coat colour is largely cosmetic, though in some breed societies it is used as a marker of breed purity.

A competent breeder does not ignore non-economic traits entirely, but ensures that selection pressure is placed overwhelmingly on economic traits, since these are what ultimately determine whether the enterprise is profitable.

1.2 Genetic Concepts

Three genetic concepts underpin all practical breeding decisions in beef and dairy production.

- Heritability is the proportion of the variation seen in a trait that is due to genetic differences between animals, as opposed to environmental differences (such as feeding or management). Traits with high heritability (such as milk fat percentage) respond well to selection, because superior parents reliably produce superior offspring; traits with

low heritability (such as fertility) are strongly influenced by environment and management, so genetic selection alone will not fix problems in these areas.

- Hybrid vigour, also called heterosis, is the improved performance often observed in crossbred animals compared with the average of their two parent breeds. Crossbreeding programmes deliberately exploit hybrid vigour to improve traits such as growth rate, fertility, and disease resistance, particularly in tropical production systems where crosses between indigenous (heat- and disease-tolerant) breeds and exotic (high-yielding) breeds are common.
- Inbreeding is the mating of closely related animals. It increases homozygosity (the proportion of gene pairs that are identical), which can fix desirable traits but far more often exposes harmful recessive genes, leading to inbreeding depression: reduced fertility, growth rate, and general vigour. Responsible breeding programmes deliberately avoid mating close relatives except in specialised line-breeding programmes managed by experienced breeders.

1.3 Selection Methods

Several methods exist for identifying superior breeding animals, each with different levels of accuracy and practicality.

- Individual merit (or mass selection) selects animals based on their own observed performance — for example, choosing the highest-yielding cows in a herd as replacement dams. It is simple and cheap but does not account for the influence of environment on the animal's own performance.
- Pedigree selection uses the recorded performance of an animal's ancestors (parents, grandparents) to predict its own genetic merit. It is useful before an animal has produced its own records (for example, selecting a young bull based on his dam's milk yield) but is less reliable than the animal's own performance once that becomes available.
- Progeny testing evaluates a breeding animal, typically a bull, based on the average performance of its offspring. Because a bull can sire hundreds or thousands of offspring through artificial insemination, progeny testing gives a very accurate picture of the genetic value the bull actually transmits, making it the gold standard for selecting dairy sires, though it is slow since it requires waiting for offspring to mature and produce records.

- Show-winning selection chooses animals based on their placing in livestock shows and competitions, judged mainly on conformation and appearance. While useful for promoting breed standards and marketing, show performance does not always correlate strongly with genuine economic merit, so it should be used alongside, not instead of, performance-based methods.

1.4 Breeding Practices

Once superior animals have been selected, breeding practices govern how mating is actually carried out.

- Mating methods include natural service, where a bull runs with the herd and mates cows as they come into heat, and artificial insemination (AI), where semen collected from a proven bull is introduced into the cow's reproductive tract by a trained technician. AI allows a farmer to access superior, progeny-tested genetics without keeping a bull, and dramatically increases the number of offspring a single superior sire can produce.
- Heat detection is essential for successful AI timing. Signs of heat (oestrus) in cattle include restlessness, mounting or standing to be mounted by other cows, a swollen and reddened vulva, mucus discharge, and reduced feed intake. Missing the heat period is one of the most common causes of poor conception rates on AI-based farms.
- Gestation in cattle lasts approximately 280 days (about 9 months), after which parturition (calving) occurs. Close observation is required around the expected calving date to assist with any difficulties and to ensure the calf begins suckling colostrum promptly.
- Weaning is the process of separating the calf from the dam's milk, transitioning it onto solid feed. The timing of weaning affects both the calf's development and the dam's return to breeding condition.

1.5 Breeding Programme

A structured breeding programme translates these individual practices into a coherent, long-term strategy. A typical 5-year herd improvement plan sets out yearly targets for the proportion of the herd bred by AI versus natural service, the culling of poor-performing animals, the introduction of new genetics, and the tracking of herd-average performance (milk yield, growth rate, calving interval) against baseline figures, so that genetic progress can be objectively measured year on year. Foundation stock the initial animals used to start or upgrade a herd —

should always be sourced from reliable, reputable breeders with verifiable performance and health records, rather than from random markets or auction houses where the health status and true genetic background of the animal cannot be confirmed.

2.0 HOUSING AND EQUIPMENT

Housing protects cattle from adverse weather, facilitates efficient management, and directly influences productivity and health. The choice of housing system depends on the scale of the enterprise, the level of intensification, climate, and available capital.

2.1 Housing Systems

- Stall barns (also called stanchion barns) confine each cow to an individual stall for most of the day, where she is fed, milked, and rested. This system allows close individual monitoring and feeding but is labour-intensive and limits the animal's freedom of movement.
- Loose housing allows cattle to move freely within a communal shed or yard, with separate areas for feeding, resting, and milking. It is best suited to extensive systems and requires less labour per animal than stall barns, though individual monitoring is more difficult.
- Free stalls (or cubicles) provide each cow with an individual resting cubicle that she can enter and leave at will, combining some of the individual comfort benefits of stall barns with the freedom of movement of loose housing; this system is now the standard in most modern commercial dairies.
- Milk parlours are specialised structures, separate from the main housing area, where cows are brought specifically to be milked, typically using mechanised milking equipment. Parlours (such as herringbone or rotary designs) allow several cows to be milked simultaneously by one operator, greatly improving milking efficiency and hygiene compared with milking cows in their stalls.

2.2 Equipment

- Feeding racks and troughs organise feed distribution, preventing wastage and contamination of feed with faeces or urine.
- Watering systems (troughs, automatic waterers) must provide constant access to clean water, since water intake is directly linked to feed intake and milk yield.

- Loading chutes are ramped structures used to safely load and unload cattle onto transport vehicles, reducing injury risk to both animals and handlers.
- Manure spreaders mechanically distribute collected manure onto pasture or cropland, turning a waste product into a valuable organic fertiliser while keeping housing areas clean.
- Milking machines mechanically extract milk via pulsating vacuum pressure applied to the teats, dramatically improving milking speed and consistency compared with hand milking, particularly in medium- to large-scale dairy operations.

2.3 Design Principles

Regardless of the specific housing system chosen, good design must prioritise adequate ventilation to remove heat, moisture, and ammonia buildup; effective drainage to keep floors dry and reduce disease and hoof problems; strict hygiene to minimise disease transmission; comfort, including adequate resting space and protection from extreme weather; and an efficient workflow that minimises the distance and time required to move animals, feed, and milk through the facility, since poor layout directly increases labour cost and animal stress.

3.0 FEEDING AND NUTRITION

Feeding and nutrition determine how much of an animal's genetic potential is actually realised. Cattle are ruminants, meaning their digestive system is specially adapted to convert fibrous plant material, indigestible to humans, into meat and milk.

3.1 The Ruminant Digestive System

The ruminant stomach has four compartments, each with a distinct function. The rumen, the largest compartment, houses a vast population of bacteria, protozoa, and fungi that ferment fibrous feed, breaking down cellulose into volatile fatty acids that the animal absorbs as its primary energy source; this microbial fermentation is what allows cattle to derive nutrition from grass and other roughages that monogastric animals cannot digest. The reticulum, closely associated with the rumen, traps heavier and foreign objects and aids in the formation of the bolus during rumination (cud-chewing). The omasum absorbs water and some nutrients from the digesta before it passes onward. The abomasum, often called the 'true stomach', functions like the stomach of a monogastric animal, secreting acid and enzymes for further chemical digestion before the digesta enters the small intestine.

3.2 Pasture and Forage

Pasture and forage form the nutritional base of most beef and dairy systems in the tropics.

- Grasses such as elephant grass (*Pennisetum purpureum*) and guinea grass (*Panicum maximum*) are the dominant forage source in many Nigerian and West African production systems, valued for their bulk yield and reasonable protein content when harvested young.
- Legumes such as alfalfa and clover are of particular value because, through their symbiotic relationship with nitrogen-fixing bacteria in their root nodules, they typically contain considerably more crude protein than grasses, and their inclusion in a pasture improves both animal performance and soil fertility.

3.3 Grazing Management

Rotational grazing divides pasture into paddocks and moves animals between them on a planned schedule, allowing grazed paddocks time to regrow before being grazed again. This practice directly prevents overgrazing — the depletion of pasture caused by allowing animals to graze a single area too intensively or for too long — which, if left unmanaged, leads to soil exposure, erosion, weed invasion, and a long-term decline in pasture productivity.

3.4 Forage Conservation

- Hay is forage that has been cut and dried (typically sun-dried) to a low moisture content (around 15–20%) for storage, allowing surplus pasture grown during the rainy season to be preserved and fed during periods of scarcity such as the dry season.
- Silage is forage that has been harvested at a higher moisture content and preserved through controlled anaerobic fermentation, in which lactic acid bacteria convert plant sugars to lactic acid, lowering the pH and preventing spoilage organisms from growing. Silage retains more of the original forage's nutritional value than hay but requires proper compaction and sealing (commonly in a silo, pit, or plastic-wrapped bale) to exclude air.

3.5 Concentrates and Rations

Concentrates are energy- and nutrient-dense feeds used to supplement forage, particularly for high-producing animals whose nutritional requirements exceed what forage alone can supply. Energy feeds such as maize and sorghum form the bulk of most concentrate mixes, supplying

the additional calories needed to support high milk yield or rapid growth. A balanced ration combines forage and concentrate in proportions calculated to meet the specific nutritional requirements energy, protein, minerals, and vitamins of a particular category of cattle, since a growing calf, a lactating cow, and a dry (non-lactating) cow all have substantially different nutritional needs.

4.0 MANAGEMENT OPERATIONS

Beyond feeding and housing, a range of routine management operations are necessary to identify animals, control disease and parasites, and manage the herd through key life-stage transitions.

4.1 Identification

- Ear tags are the most common identification method, using a numbered plastic or metal tag clipped through the ear, allowing quick visual identification of individual animals.
- Branding applies a permanent mark (traditionally with a heated iron, though freeze-branding is also used) to the animal's hide, providing identification that cannot be lost, unlike an ear tag, though it does cause some pain and hide damage.
- Tattooing places a permanent identification mark, usually inside the ear, using tattoo ink and needles; it is commonly used alongside ear tags as a backup identification method should the tag be lost.

4.2 Parasite Control

Dipping and spraying apply acaricidal (tick-killing) and insecticidal chemicals to the animal's coat, either by driving animals through a dipping tank filled with chemical solution or by directly spraying the chemical onto the animal. Regular dipping or spraying is essential in tropical regions to control external parasites such as ticks, which are not only a direct nuisance but also transmit serious diseases including babesiosis and anaplasmosis.

4.3 Physical Operations

- Dehorning removes or prevents the growth of horns, primarily to prevent injuries to other animals and to handlers during close contact such as milking.
- Castration, the removal or destruction of the testicles in male animals not intended for breeding, improves meat quality (reducing the strong odour and flavour associated with

intact bulls) and produces calmer, easier-to-manage animals; it is also, by definition, a method of controlling uncontrolled breeding within the herd.

- Weighing at regular intervals monitors growth rate, allowing early detection of animals that are falling behind expected growth curves and informing decisions on feeding adjustments, marketing timing, and breeding readiness.

4.4 Cow-Calf Management

Cow-calf management covers the entire period from calving through to weaning, encompassing appropriate feeding of both dam and calf, disease prevention (including ensuring the calf receives adequate colostrum within the first few hours of life for passive immunity), careful record keeping of calving dates, calf identification, and growth, and a planned weaning process that minimises stress on both cow and calf.

5.0 MILKING AND MILK HANDLING

For dairy enterprises, the milking process and subsequent milk handling determine both the quantity of milk captured and its quality and safety for sale or processing.

5.1 Milk and Milk Products

Beyond fresh liquid milk, dairy processing yields a range of products including butter (produced by churning cream to separate the fat), cheese (produced by coagulating milk proteins, typically using rennet or acid, and separating the resulting curd from the whey), and yoghurt (produced by fermenting milk with specific bacterial cultures that convert lactose to lactic acid, thickening the milk and giving it its characteristic tangy flavour).

5.2 Udder Anatomy

The bovine udder is divided into four independent quarters, each with its own teat, milk-producing tissue, and duct system, meaning an infection or injury in one quarter does not necessarily spread to the others. Milk let-down the release of milk from the udder's storage tissue into the ducts, making it available for milking is triggered by the hormone oxytocin, released in response to the calf suckling or, in a milking parlour, in response to the conditioned stimuli associated with the milking routine (such as udder washing and the sound of the milking machine).

5.3 Milking Techniques

- Hand milking manually expresses milk by rhythmic compression of the teat between fingers and palm; it remains common on smallholder farms but is slower and more labour-intensive than machine milking.
- Machine milking uses a mechanised system that applies intermittent vacuum pressure to the teat, mimicking the suckling action of a calf, and is standard practice on medium- to large-scale commercial dairies due to its speed, consistency, and reduced labour requirement.

5.4 Factors Affecting Milk Yield

Milk yield is influenced by nutrition (an under-fed cow cannot produce to her genetic potential), health (mastitis and other diseases directly suppress yield), and environment (heat stress in particular is a major depressor of milk yield in tropical climates, since cattle divert energy to thermoregulation rather than milk synthesis).

5.5 Milk Handling

Accurate recording of milk yield per cow, per milking, is essential both for monitoring individual animal performance and for informing culling and breeding decisions. Once collected, milk intended for sale or further processing typically undergoes pasteurisation (controlled heating to destroy pathogenic bacteria while preserving nutritional quality) and, for some products, homogenisation (mechanically breaking down fat globules so they remain evenly suspended rather than separating into a cream layer). Prompt refrigeration after milking and throughout the supply chain is critical, since milk is an excellent growth medium for bacteria and spoils rapidly at ambient tropical temperatures.

6.0 DISEASES AND PARASITES

Disease and parasite management protects both animal welfare and farm profitability, since sick animals produce less milk and meat, may require costly treatment, and in some cases pose a risk to human health or the wider herd.

6.1 Classification of Diseases

- Bacterial diseases, including anthrax and brucellosis, are caused by bacterial infection and are often treatable with antibiotics if caught early, though some (such as anthrax) can be rapidly fatal and pose serious zoonotic risk to humans.

- Viral diseases, such as foot-and-mouth disease, have no direct antibiotic cure and are managed primarily through vaccination and strict biosecurity, since the disease spreads highly efficiently between cloven-hoofed animals.
- Protozoan diseases, such as trypanosomiasis (transmitted by the tsetse fly), are caused by single-celled parasites and are a major constraint on cattle production across large parts of tropical Africa.
- Nutritional disorders, such as milk fever (hypocalcaemia, a drop in blood calcium typically occurring around calving) and ketosis (an energy deficit disorder in early lactation), arise from metabolic imbalances rather than infectious agents and are managed through targeted nutritional intervention around key physiological transitions.
- External and internal parasites, including ticks and worms, cause direct damage (blood loss, hide damage, reduced weight gain) and, in the case of ticks, transmit further infectious diseases.

6.2 Prevention

Disease prevention rests on three pillars: hygiene (clean housing, water, and equipment to reduce pathogen load), vaccination (stimulating the animal's immune system in advance of exposure to specific pathogens), and sound general management (adequate nutrition, reduced stress, and biosecurity measures such as quarantining newly introduced animals).

6.3 Vaccination Programmes

An effective herd health plan follows a scheduled immunisation programme, timed to key life stages (such as pre-calving or weaning) and to the seasonal risk periods of specific diseases prevalent in the local area, rather than vaccinating reactively only after disease has already appeared in the herd.

6.4 Control

Once disease or parasite infestation is identified, control relies on appropriate drug therapy (antibiotics for bacterial infections, antiparasitics for worm or tick burdens), combined with correcting any underlying housing or management deficiencies such as poor drainage or overcrowding that predisposed the herd to the problem in the first place, since treating the immediate case without addressing the underlying cause invites recurrence.

7.0 RECORD KEEPING AND MARKETING

A beef or dairy enterprise cannot be efficiently managed, or profitably sold into, without accurate records and a clear understanding of the market.

7.1 Farm Records

- Breeding records document mating dates, sire used, expected and actual calving dates, and calving difficulty, forming the basis for all future selection and breeding decisions.
- Milking records track daily or periodic yield per cow, enabling early identification of both top performers (for breeding) and declining performers (for veterinary attention or culling).
- Accounting records capture all income and expenditure associated with the enterprise, from feed and veterinary costs to the sale of milk, calves, or cull animals.

7.2 Importance of Record Keeping

Good records directly support day-to-day decision-making (such as which cow to breed next or which animal to cull), enable ongoing monitoring of productivity trends against targets, and underpin sound financial planning, since a farmer who does not know the actual cost of producing a litre of milk or a kilogram of beef cannot reliably judge whether the enterprise is profitable.

7.3 Marketing

Successful marketing begins with correctly identifying suitable cattle markets whether local live markets, abattoirs, cooperative milk collection centres, or direct sale to processors and understanding their specific requirements and price structures. Buyers typically assess animals against standard conformation traits: body shape, muscling, frame size, and condition score, which serve as visual proxies for expected carcass yield and quality in beef animals, or general health and productive capacity in dairy animals. A farmer who understands these conformation standards is better positioned to both select breeding stock that will meet market expectations and negotiate fair prices when buying or selling animals.

Milk

Milk is a creamy yellowish liquid formed in the mammary glands of mammal and is used by all female mammals for feeding their young ones from a period beginning immediately after birth. Milk can be produced from cows, horses, goats, ewes, monkeys, dogs, sheep etc but cow's milk is the most commonly used worldwide. Dairy or milk products may be defined as all foods and other commodities derived from milk. About 75% of the milk produced in animal husbandry is used in its liquid form, and most consumer milk is pasteurized. Usually, milk is also homogenized. Milk contains a wide range of nutrients with a balance macronutrients (4.9% carbohydrates, 3.7% fats and 3.5% proteins) composition. It is also consist of about 87.20% water.

(A) **Carbohydrate:** The type found in milk is lactose which is often called milk sugar. Lactose is digested by the enzyme lactase into simpler sugar that the body can absorb.

(B) **Fat:** The fat portion of milk, called the butter, is in form of minute globules which tend to rise to the milk to form a cream line. The fatty acids in milk are mostly of the saturated and mono-saturated types, although there are also small amount of poly unsaturated fatty acids in milk. The minimum content acceptable worldwide is 3.25%

(C) **Protein:** The protein in milk is of high quality. They are rich in all the essential amino acids. They are casein, lacto albumen and lacto globulin.

(D) **Vitamins:** Milk contains both water-soluble vitamins e.g. thiamine, B1, riboflavin, B2, B12 and fat soluble vitamins such as A,D,E, and K. Milk also contains carotene which is responsible for the yellowish colour of milk at times.

(E) **Minerals:** Milk contains all the trace elements known to be important to the body. E.g. zinc, iodine, selenium and chromium but relatively poor source of iron, copper and manganese. Milk is rich in some essential mineral elements such as calcium and phosphorus.

TYPES OF MILK

1. Raw milk: This is an unpasteurized milk straight from cow.
2. Fresh whole milk: This is the milk that is obtained directly from the cow in which none of the nutrients have been removed.
2. Skimmed milk: The fat content (or cream) has been removed, and it is therefore made predominantly of protein, carbohydrate, minerals and vitamins.
3. Evaporated milk: This is whole from which about 60% of the water content has been removed.
4. Dried or powdered milk: Over 90% of the water content has been removed and it is then milled to powdered form.

5. Condensed milk: This is evaporated milk to which a safe and suitable nutritive sweetener usually sugar has been added. So it is sweeter and thicker than evaporated milk.

Methods of processing milk

(A) PASTEURIZATION

Milk is heated for a short time (about 30 minutes) at a temperature below 100°C (nearly 70°C) to kill the harmful bacteria present in it. High temperature, short time (HTST) method is being used nowadays, i.e. milk is heated quickly to 71°C, held at the temperature for 15 seconds and cooled rapidly.

(B) HOMOGENIZATION

Milk is first pasteurized and then forced through a fine aperture under pressure. This breaks the fat globules, completely that a cream line is formed. The fat remains evenly distributed throughout the milk to obtain a homogenous fluid. This is usually achieved by passing the whole fresh milk through a very tiny nozzle in a special piece of equipment known as homogenizer.

(C) STERILIZATION

Milk is pre-heated and homogenized before filling into bottles which are closed in an air-tight seal (hermetically sealed). The bottles are then heated to 104°C – 110°C and kept at this temperature for 20 to 30 minutes. Sterilized milk contains no bacteria and the bottles are vacuum sealed and can thus be kept for several weeks without opening. The high temperature used in sterilization causes slight caramelization of the lactose producing a noticeable change in flavour and appearance. There is also reduction in the nutritive value with loss of one third of the thiamine (vitamin B1) and half of the ascorbic acid (vitamin C), folic acid and vitamin B12. Sterilized milk is a good standby; with its creamy taste, it is ideal for puddings.

(D) EVAPORATED MILK: This is homogenized and concentrated (by removal of about 60% of its water content) to about half the volume of the original milk without adding sugar. It is sterilized in the can and some of the thiamine (B1) and vitamin C are destroyed

(E) CONDENSED MILK: It is also a concentrated form of milk. It is made from whole skimmed or semi-skimmed milk with sugar added.

(F) DRIED MILK: Over 90% of the water content has been removed and it is then milled to powder. The ranges of dried milk available include skimmed milk, skimmed milk to which vegetable fat has been added known as filled milk.

(G) CULTURED OR FERMENTED MILK: Milk may be fermented with/of products specific microorganism to form a variety with characteristic level of acidity and flavour

development. The best known milk produce is yoghurt. Others are cultured butter milk, 'nono', etc.

MILK PRODUCTS

Milk products can be broadly classified into fermented and non-fermented products. Examples of fermented milk products are yoghurt, sour milk curds, cheese and local chesses called "wara". Milk products that are not fermented are butter, caesinates and milk ice-cream.

(a) Yoghurt

This is obtained from heat-treated, homogenized milk which is inoculated with a culture containing equal amount of *Streptococcus thermophilus* and *Lactobacillus bulgaricus* bacteria. The essential change produced by these bacteria is that the lactose in milk is converted to lactic acid.

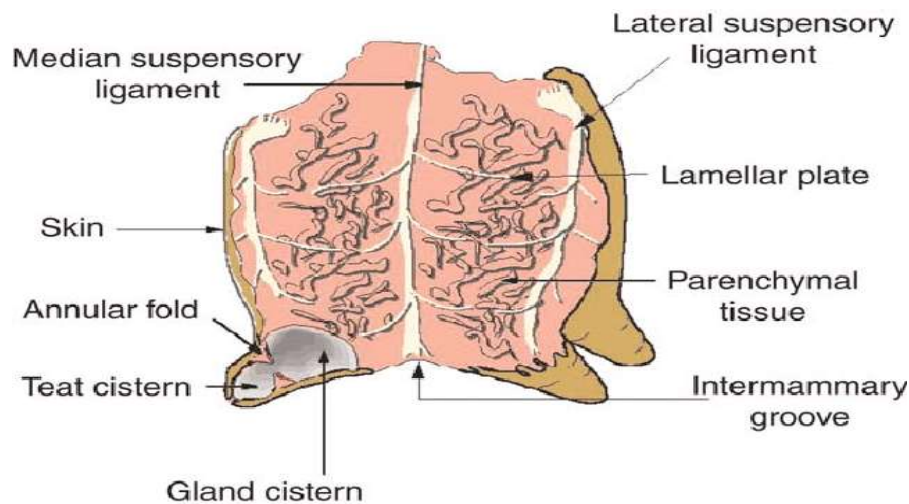
(b) Cheese

There are more than 800 different types of cheese; the basic principles of manufacturing are the same. Milk is coagulated and the solid formed is cut into small pieces to allow the whey (the liquid part of the milk) to drain off. The curd (solid part of the milk) is dried, salt is added and the cheese and texture of the cheese is lightly pressed, a soft crumby cheese is produced but it is more firmly pressed, a harder cheese is produced.

Cheese types include Soft unripened (e.g. cottage chesses) ripened, (e.g. camembert), Semisoft (e.g. Munster, limburg, Roquefort, gorgonzola stallion, wenslaydale etc), Hard (e.g. cheddar, granular, Swiss, emmentaler, gruyere, caciocavallo), Very hard (e.g. parmesan, Romano, sapsago, spleen, etc), Whey cheese (e.g. my sost, primo's, ricotta etc)

UDDER

An udder is a large, hanging skin organ located in the groin or inguinal area of female dairy animals and ruminants like cows, goats, and sheep that produces and holds milk. It features four distinct quarters (in cows) with protruding teats, and an internal network of milk-producing alveoli and storage cisterns. **Mammary glands** produce milk using raw materials filtered from the animal's blood supply. Cows typically have four glands and four teats, while sheep and goats generally have two glands and two teats.



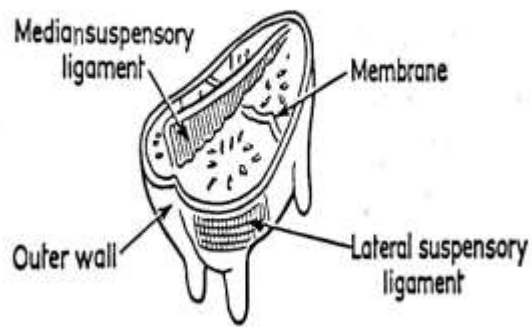
Structure of Udder

External Features

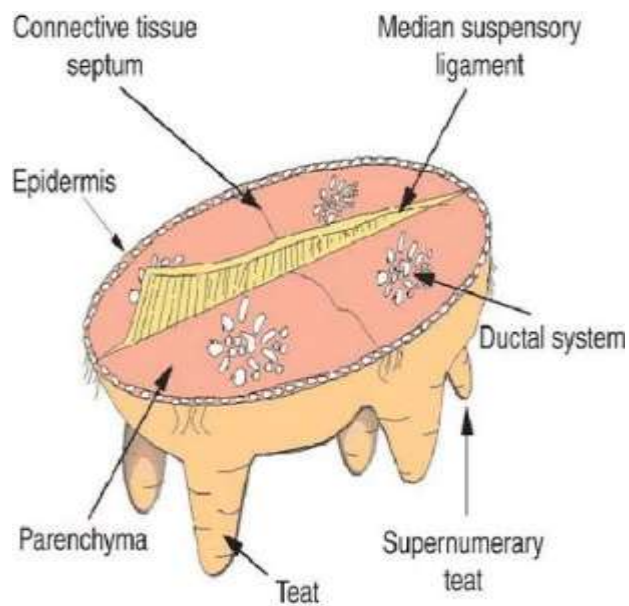
1. **Halves and Quarters:** A cow udder has two distinct sides (left and right) and four functional glandular units called quarters (two fore and two rear). Smaller ruminants like sheep and goats feature two quarters and two teats. Rear quarters are often larger and hold more milk than front quarters.
2. **Intermammary Groove:** A visible external median line or cleft separating the left and right halves of the udder from front to rear.
3. **Floor:** The flat or slightly rounded bottom surface where the teats attach.
4. **Skin Coverage:** The main body of the udder is covered in soft, pigmented skin with fine hairs, which protects the inner tissues.
5. **Teats (Nipples):** Cylindrical, hair-free protruding projections hanging from each quarter. Each teat features a smooth outer surface when full and a wrinkled texture after milking, ending in a tiny opening (orifice) that lets milk out.
6. **Supernumerary Teats:** Extra, non-functional small teats sometimes present near the main ones

Internal Structure

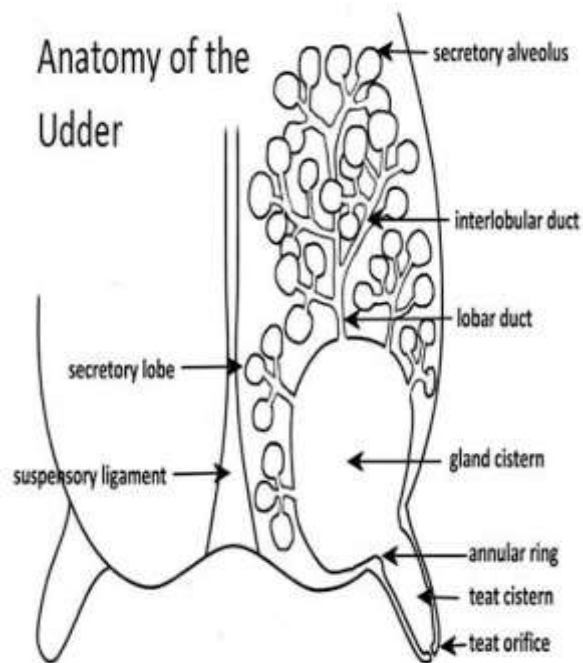
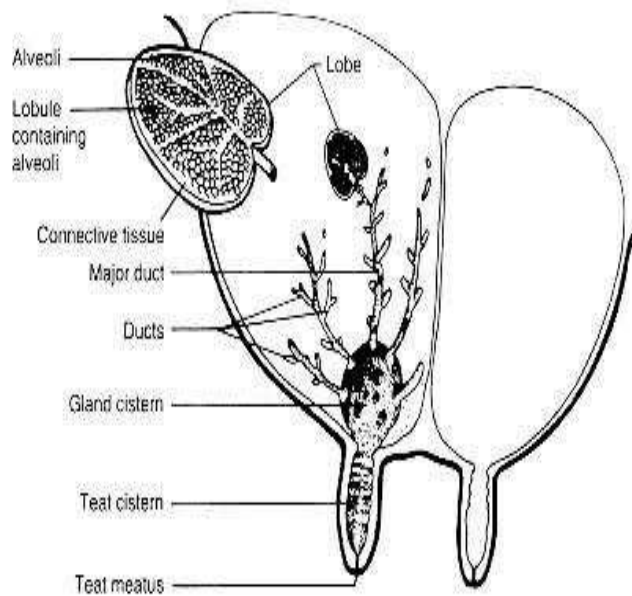
The internal structure of an udder consists of four independent functional quarters, containing milk-producing **alveoli**, a network of **milk ducts**, and storage **cisterns**. Each quarter operates as a separate gland with its own teat. The milk which is continuously synthesized in the alveolar area is stored in the alveoli, milk ducts, udder and teat cistern between milkings. 60-80% of milk is stored in the alveoli and small milk ducts, while the cistern only contains 20-40%.



Cross section of udder, showing the four separate quarters.



Cross section of the four quarters of the udder



Anatomy of the udder

Glandular and Secretory Tissue (Parenchyma)

1. Alveoli: Microscopic, hollow, grape-like clusters where milk is actively synthesized from blood components. Each alveolus is lined with a single layer of secretory epithelial cells and wrapped in myoepithelial muscle cells that contract to force milk out. The diameter of each alveoli is about 50 – 250 μ m.

2. **Lobules and Lobes:** Groups of 150 to 220 alveoli bound together by connective tissue form a lobule, and collections of lobules form a lobe

Duct and Storage System

3. **Milk Ducts:** Progressively larger tubes that collect milk from the alveoli and transport it downward
4. **Gland Cistern (Milk Cistern):** A larger cavity at the base of each quarter that holds between 100 to 450 mL of accumulated milk before milking or until milk letdown occurs.
5. **Teat Cistern:** A smaller cavity located inside the teat directly beneath the gland cistern. It temporarily stores milk right before it leaves.
6. **Streak Canal (Teat Canal):** The external opening channel at the base of the teat regulated by a muscular sphincter muscle that keeps milk from leaking and blocks bacteria (due to the presence of keratin).
7. **Rosette of Furstenberg:** 6-10 longitudinal folds of mucous membrane located at the junction of the teat cistern and streak canal. It helps to keep milk in and stop bacteria from entering.

Support and Vascular Framework

8. **Median Suspensory Ligament:** A strong, elastic central ligament containing yellow elastic tissue that divides the left and right halves and supports most of the udder's weight
9. **Lateral Suspensory Ligaments:** Inelastic, fibrous sheets wrapping around the outer sides of the udder to anchor it to the pelvic and abdominal tendons.
10. **Vascular and Nerve Supply:** A dense network of arteries, veins (such as the subcutaneous abdominal "milk vein"), and nerves supplying the extreme volume of blood required to synthesize milk

Milk letdown

The contraction of the alveoli and expulsion of alveolar milk into the udder cistern is called milk letdown. Milk let-down in ruminants is a neuroendocrine reflex driven by oxytocin release, myoepithelial contraction, and sensory stimulation. It is the vital process that moves milk from the tiny storage sacs (alveoli) inside the udder down into the teat cisterns so it can be removed. It has to occur before a cow will *milk* out freely

The Process of Milk Let-down or Ejection

1. **Stimulation:** The young animal nosing the udder, or a farmer washing and touching the teats, sends nerve signals to the brain.

2. **Hormone Release:** The brain releases the hormone oxytocin into the blood.
3. **Contraction:** Oxytocin travels to the udder and forces small muscle cells (myoepithelial cells) around the milk sacs to squeeze.
4. **Flow:** Squeezing pushes the milk out into the large ducts and teats, making it ready for milking or suckling. This takes about 30 seconds to 2 minutes to start and lasts for roughly 5 minutes

Techniques for milking cow

The techniques for milking cow can be broadly classified into two: Manual milking techniques and Mechanical and automated techniques.

1. Manual Milking Techniques

Cows are milked from left side. after let down of milk, the milker starts milking teats either cross wise or fore quarters together and then hind quarters together or teats appearing most distended milked first few streams of fore milk from each teat be let on to a strip cup. This removes any dirt from the teat canal and gives the operator a chance to detect mastitis.



A. **Full-Hand Milking (Fisting):** Grasp the entire teat in your hand. Pinch the top of the teat with your thumb and index finger to trap the milk. Squeeze sequentially down with your remaining fingers (middle, ring, pinky) to force the milk out in a steady stream. The teat is compressed and relaxed alternatively in quick succession, thus the method removes milk much quicker than stripping as there is no loss of time in changing the position of the hand. Also, full hand method is superior to stripping as it stimulates the natural suckling process by calf and moreover the method exerts an equal pressure on the large teats of cows and buffaloes. Hence, it is best for cows or buffaloes with large teats.

B. **Stripping:** Hold the base of the teat tightly between your thumb and forefinger. Slide your fingers down the full length of the teat while maintaining pressure to push the milk out. This is best used at the end of milking to empty residual milk or for cows with small teats.

C. **Knuckling (Discouraged):** This involves pressing the teat against the inner edge of the thumb and bent knuckles. This technique is **not recommended** because it bruises and damages sensitive tissue inside the teat.

2. Mechanical and Automated Techniques

Mechanical and automated milking techniques save time and energy, facilitate removal of all the milk from udder, adaptable, give a suckling feeling to the cow and avoid pain in the udder as well as leakage of milk.

A. **Machine Milking:** This involves attaching rubber or silicone teat cups to each of the four teats. It uses a periodic vacuum and air pressure pulsation cycle (typically 48–72 cycles per minute) to gently massage the teat and extract milk cleanly and efficiently. The milking machine works based on two basic principles: 1. It opens the streak canal through the use of a partial vacuum, allowing the milk to flow out of the teat cistern through a line to a receiving container. 2. It massages the teat, which prevents congestion of blood and lymph in the teat.



B. **Robotic Milking:** Cows voluntarily enter an automated stall where sensors locate teats and complete the cleaning, milking, and post-dip process without human intervention.

Best Hygiene Practices

1. Clean and dry the udder with warm water or pre-dips before starting.

2. Perform forestripping (drawing the first few streams into a separate cup) to check for clots or signs of mastitis.
3. Apply a post-milking antiseptic teat dip to seal the teat canal against bacteria

Factors affecting the quantity and quality of milk

1. **Breed and Species:** Different animal species and breeds (such as Holstein-Friesian vs. local breeds) determine baseline milk yield and fat-to-protein ratios. Also, buffalo milk contains more milk fat than cow milk.
2. **Stage of Lactation:** Yield peaks a few weeks after calving and drops over time. High content of fat and protein is found in the initial stage of lactation. As lactation progresses and milk yield increases, there is rapid reduction in fat and protein contents and an increase in lactose content. The size of fat globules is bigger during the first few months and it remains fairly constant during first, sixth and seventh month of lactation and again becomes smaller during the last month of lactation.
3. **Age:** Milk yield in cows continues increasing till 9-10 years of age and then starts declining. Protein and fat declines with age.
4. **Health Status:** Diseases like mastitis reduce total volume, alter chemical composition, and lower overall milk grade.
5. **Feed and Nutrition:** The type, quality, and balance of forage and concentrates directly dictate milk fat, protein levels, and total daily output.
6. **Climatic Conditions:** High ambient temperatures and humidity cause heat stress, decreasing feed intake, milk volume, and solid components.
7. **Milking and Hygiene:** Regular milking frequency, clean handling procedures, and proper storage prevent bacterial contamination and preserve raw milk quality.

Milk production record

A milk production record tracks daily or weekly milk yields from individual dairy animals to help farmers manage feeding, health, and breeding. To prepare a dairy milk production record, you need to set up a log system, capture daily milking volumes, and track individual animal data. The record should contain animal ID (tag number or name), date and lactation stage (calendar date and days in milk), milking sessions and yield, daily total (sessions addition for each animal), remarks (Note any health issues, feed changes, or veterinary treatments) and monthly summaries.

Processing and storing milk

Effective processing and storage of milk involves straining, pasteurization, cooling, and keeping it sealed in a cold refrigerator.

Straining: Pour the fresh milk through a clean cloth or a specialized dairy filter right after milking to catch any dirt or dust.

Pasteurization: Heat the milk to kill bad germs. Heat it to 161°F (72°C) for 15 seconds or 145°F (63°C) for 30 minutes using a double boiler, making sure not to let it boil.

Cooling: Place your container of hot milk into an ice-water bath and stir it gently until the temperature drops fast.

Storage: Use clean, sanitized glass jars or food-grade plastic containers with tight lids to store the milk in a refrigerator at **36°F to 40°F (2°C to 4°C)**.